PUNTITA SAE-JIA

088-1971271 puntita.jia@gmail.com

Education

- Ph.D. in Mathematics (International Program), Chiang Mai University, Thailand, 2026
  Dissertation defense completed; awaiting final approval from the University Council.
- M.Sc. in Mathematics, Chiang Mai University, Thailand, 2023
- B.Sc. in Applied Mathematics (First Class Honors), King Mongkut's Institute of Technology Ladkrabang (KMITL), Thailand, 2020

Languages

- Thai – Native
- English – Working proficiency

Technical Skills

- Programming: Python, MATLAB, LaTeX
- Automation & Development: n8n, Web Development
- Software: Microsoft Word, Excel, PowerPoint, Keynote

Reference

- **Prof. Dr. Suthep Suantai**
  Chiang Mai University
  Telephone: 053-943326 ext. 138
- **Prof. Dr. Piyapong Niamsup**
  Chiang Mai University
  Telephone: 053-943326 ext. 130
- **Prof. Dr. Bancha Panyanak**
  Chiang Mai University
  Telephone: 053-943326 ext. 134
- **Prof. Dr. Yeol Je Cho**
  Gyeongsang National University, Republic of Korea
  Telephone: +82 55-772-2785
- **Prof. Dr. Jen-Chih Yao**
  National Sun Yat-sen University, Taiwan
  Telephone: +886 7-335-6047

Research Interests

- Machine Learning
- Optimization Methods for Artificial IntelligenceFixed Point Theory
- Bilevel Optimization and Variational Inclusion Problems
- Health Data Analytics and Artificial Intelligence in Medicine
- Predictive Modeling for Healthcare Systems and Policy

Google Scholar & Citation Profile

Google Scholar: H-index 2 Citations 9
Total Publications: 5

Publications

- Sae-jia, P., and Suantai, S., "A Novel Algorithm for Convex Bi-level Optimization Problems in Hilbert Spaces with Applications," Thai Journal of Mathematics, Vol. 21, No. 3, 2023, pp. 625–645.
  Quartile: Q3
- Sae-jia, P., and Suantai, S., "A New Two-Step Inertial Algorithm for Solving Convex Bilevel Optimization Problems with Application in Data Classification Problems," AIMS Mathematics, Vol. 9, No. 4, 2024, pp. 8476–8496. DOI: https://doi.org/10.3934/math.2024412
  Quartile: Q1
- Sae-jia, P., and Suantai, S., "A New Two-Step Inertial Algorithm for Solving Convex Bilevel Optimization Problems with Application in Data Classification Problems," AIMS Mathematics, Vol. 9, No. 4, 2024, pp. 8476–8496. DOI: https://doi.org/10.3934/math.2024412
  Quartile: Q1
- Sae-jia, P., Panyahan, E., and Suantai, S., "A New Accelerated Forward–Backward Splitting Algorithm for Monotone Inclusions with Application to Data Classification," Mathematics, Vol. 13, No. 17, 2025, Article 2783. DOI: https://doi.org/10.3390/math13172783
  Quartile: Q1
- Sae-jia, P., and Suantai, S., "A Double Inertial Fixed Point Algorithm with Linesearch and Its Application to Machine Learning for Data Classification," Carpathian Journal of Mathematics, Vol. 42, No. 2, 2026, pp. 199–216. DOI: https://doi.org/10.37193/CJM.2026.02.02
  Quartile: Q1

Manuscript Under Review

- Sae-jia, P., Panyahan, E., Suantai, S., and Yao, J.-C., "Double Inertial Viscosity Forward-Backward Algorithm for Split Variational Inclusion with Applications to Deblurring of Medical Images and Data Classification," Under review at Journal of Computational and Applied Mathematics.

Conference Presentations

- **ICWANA 2024**

  Accelerated Algorithms for Convex Bilevel Optimization Problems with Applications for Data Classification Using Extreme Learning Machine Model (6–8 August 2024, Thailand)

- **ICNS 2024**

  Accelerated Common Fixed Point Algorithms Using Viscosity Approximation and Double Inertial Techniques for Convex Bilevel Optimization and Data Classification with ELM Models (26–28 September 2024, Japan)

- **ACFPTO 2025**

  Efficient Common Fixed Point Methods Using Viscosity Approximation and Double Inertial Techniques for Convex Bilevel Optimization and Data Classification (15–18 January 2025, Thailand)

- **AMC 2025**

  A Double-Inertial Fixed Point Algorithm for Convex Bilevel Optimization and Data Classification (3–7 August 2025, Thailand)

International Research Experience

- 23–30 September 2024

Professor Tamaki Tanaka

Niigata University, Japan

- 8 November 2025 – 6 January 2026

Professor Jen-Chih Yao

National Sun Yat-sen University, Taiwan

Research Projects and Innovation

Hypertension Predictor – Web application for hypertension risk prediction (copyright pending).

Research Experience

Research Assistant (Ph.D. level),
Excellence Center in Optimization and Computational Intelligence for Big Data Prediction

Research Assistant

Mathematical Modeling and Intelligence Algorithms Development Using Machine Learning for Non-Communicable Disease Prevention and Screening (FF66–FF68).

Research Assistant

Mathematical Modeling and Intelligence Algorithms Development Using Machine Learning for Non-Communicable Disease Prevention and Screening (FF65).

Teaching Experience

2021–Present	Gifted Mathematics Camp Instructor
2021	Teacher Assistant (Algebra)
2020–2021	Mathematics Teacher for Grade 4 – Grade 6 at Sriwarakarn School
	Computer Science Teacher for Grade 4 – Grade 6 at Sriwarakarn School
	Statistics Teacher for first year Vocational Certificate student and first year High Vocational Certificate student at Sriwarakarn Technology College
2016 - 2021	Private Mathematics Tutor (Primary – High School)

Invited Talks and Academic Services

- Invited Speaker, Training Program for Village Health Volunteers (VHVs), San Pa Tong District, Thailand. Topic: Mobile Innovation for Hypertension Risk Prediction (Hypertension Predictor). (Organized by Excellence Center in Optimization and Computational Intelligence for Big Data Prediction, Chiang Mai university)

- Invited Speaker, Workshop on Artificial Intelligence Applications to Improve Workforce Skills and Efficiency, MSP Interfood Co., Ltd. (Organized by Professor Suthep Suantai)

- Invited Speaker, Workshop on “Using AI as an Assistant for Developing Textbooks and Teaching Materials.” (Organized by Faculty of Science and Technology, Uttaradit Rajabhat University)

- Invited Speaker, Project on Mathematical Modeling and AI Algorithms Using Machine Learning for NCD Prevention and Screening. Topic: Mobile Innovation for Hypertension Risk Prediction. (Organized by Professor Suthep Suantai)

- Conference Assistant, ACFPTO 2025 and AMC 2025 International Conferences.